CS

Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CSOE02]

FIFTH /SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

PYTHON PROGRAMMING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Describe the different data types using variables and literal constants with python program.	**08**
	b.	Write the python program to Exchange the value of two variables(swapping) with and without using temporary variable.	**06**
2.	**a.**	Illuminate string handling function with examples.	**08**
	b.	What is module? How many ways to import module in python?	**06**
3.	**a.**	Using the functions and methods, analyze the differences and similarities of lists and tuples with examples for each.	**09**
	b.	Explain the significant features of a dictionary in python with Example.	05
		OR	
4.	**a.**	What are the basic list operations that can be performed in Python? Explain each operation with its syntax and example.	**07**
	b.	Explain the concept of Dictionary? Explain Python dictionaries in detail discussing its operations and methods.	**07**
5.	**a.**	What are packages? Give an example of package creation in Python. Demonstrate the use of Exception Handling in Python.	**08**
	b.	Discuss the different modes for opening a file and closing a file.	**06**
		OR	
6.	**a.**	Discover a program to catch a divide by zero exception. Add a finally block too.	**04**
	b.	Write a function that reads a file file1 and evaluates and displays number of words and vowels in the file	
7.	**a.**	Describe numpy array with an example.	**07**
	b.	How to perform linear algebra using SCIPY? Explain with examples.	**07**
		OR	
8.	**a.**	What is broadcasting in numpy? Explain with an example.	**07**
	b.	Briefly explain Scipy with an example.	**07**